

Machine Learning II - Structured Models

Exercise sheet **Coordinate descent algorithms: Primal descent (ICM) vs. Dual ascent (Diffusion).**

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Exercise 1 (Color Segmentation Problem). Download and unpack `color-seg-n4` datasets from the OpenGM benchmark <http://hci.iwr.uni-heidelberg.de/opengm2/> (or use the downloaded ones from the previous exercises). Use the python scripts to the exercise **Dual Of LP Relaxation. Reparametrization.**

ICM - primal coordinate descent ICM works in the primal domain with the non-relaxed energy function. It fixes labels assigned to all nodes except one and optimizes over all possible labels in this node. The process is repeated iteratively until none of the labels change.

Run the ICM solver on 4 – 5 datasets (e.g. `lake`, `crops`, `pfau`,...). Since the algorithm makes a lot of very small updates (for each node) output only each 100th update:

```
opengm_min_sum -a ICM -p 100 -v -o crops-icm.h5 -m crops-small.h5
```

Use the naïve solver to obtain the naïve value and bound for all datasets, e.g.:

```
python naive_solver.py crops-small.h5
```

Compare the naïve value and bound to the value of returned by the ICM.

Diffusion - dual coordinate ascent Diffusion optimizes the dual MAP LP and performs local coordinate ascent steps. On each step it updates dual variables related to a single node. Primal integer labeling is reconstructed based on the dual rounding.

Run the diffusion solver on 4 – 5 datasets (e.g. `lake`, `crops`, `pfau`,...). The solver's implementation is quite slow, however it is recommended to run it at least 100 iterations.

```
opengm_min_sum -a TRWSi -p 1 -v --maxIt 100 --d EDGE -o crops-dif.h5 -m crops-small.h5
```

Check that already after the first iteration results obtained by the diffusion algorithm are better than the final ones of the ICM. Run the `diffusion` again at least on the `crops` and `lake` datasets until the absolute duality gap is less than 0.05 :

```
opengm_min_sum -a TRWSi -p 1 -v --maxIt 1000 --d EDGE -o crops-dif.h5 -m crops-small.h5 --precision 0.05
opengm_min_sum -a TRWSi -p 1 -v --maxIt 1000 --d EDGE -o lake-dif.h5 -m lake-small.h5 --precision 0.05
```

Compare the obtained values/bounds to those obtained in the exercise **Dual Of LP Relaxation. Reparametrization..**

Exercise 2 (Stereo Reconstruction). Repeat similar experiments with `tsu-gm` instance from the `mrf-stereo` dataset:

ICM - primal coordinate descent Run:

```
opengm_min_sum -a ICM -p 100 -v -o tsu-icm.h5 -m tsu-gm.h5
python naive_solver.py tsu-gm.h5
```

Diffusion - dual coordinate ascent Run:

```
opengm_min_sum -a TRWSi -p 1 -v --maxIt 100 --d EDGE -o tsu-dif.h5 -m tsu-gm.h5
```

Check that already after the first iteration the result obtained by the **diffusion** algorithm is better than the final one of the ICM and it gets significantly better after 100 iterations. Compare the obtained values/bounds to those obtained in the exercise **Dual Of LP Relaxation. Reparametrization.**

```
opengm_min_sum -a TRWSi -p 1 -v --maxIt 1000 --d EDGE -o crops-dif.h5 -m crops-small.h5 --precision 0.05
opengm_min_sum -a TRWSi -p 1 -v --maxIt 1000 --d EDGE -o lake-dif.h5 -m lake-small.h5 --precision 0.05
```

Check how the results differ visually:

```
python show_labeling.py -V 288 -H 384 -v -C -f tsu-icm.h5
python show_labeling.py -V 288 -H 384 -v -C -f tsu-dif.h5
```

Fill out the result table and write your conclusions from the exercises.